1. Considere a proposição composta: $(P \wedge Q) \leftrightarrow (\neg P \vee R) \rightarrow Q$
Construir a tabela verdade completa para a proposição. Lembrando que o número de linhas da tabela (2^n) depende do número de proposições simples (n) na proposição composta. (2,0)

P	Q	R	~P	~P V r	P ∧ Q	P∧Q <-> (~P V r)	P ∧ Q <-> (~P V R) -> Q
V	V	V	F	V	V	V	V
V	V	F	F	F	V	F	V
V	F	V	F	V	F	F	V
V	F	F	F	F	F	V	F
F	V	V	V	V	F	F	V
F	V	F	V	V	F	F	V
F	F	V	V	V	F	F	V
F	F	F	V	V	F	F	V
					F		V

2. Operações com Matrizes, dadas as Matrizes A, B e C, calcule;

$$A = \begin{pmatrix} 2 & -1 & 0 \\ 3 & 1 & 4 \\ -2 & 5 & 3 \end{pmatrix} B = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ -2 & 3 & 1 \end{pmatrix} C = \begin{pmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ -1 & 0 & 1 \end{pmatrix}$$

a) $A + B - C$ (2,0)
b) $3.B$(2,0)
c) $A\,.\,B$ (2,0)

a) $A+B = \begin{bmatrix} 2 & -1 & 0 \\ 3 & 1 & 4 \\ -2 & 5 & 3 \end{bmatrix} + \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ -2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 & 1 \\ 3 & 2 & 6 \\ -4 & 8 & 4 \end{bmatrix}$

$A+B-C = \begin{bmatrix} 3 & -2 & 1 \\ 3 & 2 & 6 \\ -4 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -3 & -1 \\ -1 & 0 & 2 \\ -5 & 8 & 3 \end{bmatrix}$

b) $3 \times \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ -2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -3 & 3 \\ 0 & 3 & 6 \\ -6 & 9 & 3 \end{bmatrix}$

c) $A \times B = \begin{bmatrix} 2 & -1 & 0 \\ 3 & 1 & 4 \\ -2 & 5 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ -2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 2\times1+(-1\times0)+0\times(-2) & 2\times(-1)+(-1)\times1+0\times3 & 2\times1+(-1)\times2+0\times1 \\ 3\times1+1\times0+4\times(-2) & 3\times(-1)+1\times1+4\times3 & 3\times1+1\times2+4\times1 \\ -2\times1+5\times0+3\times(-1) & -2\times(-1)+5\times1+3\times3 & -2\times1+5\times2+3\times1 \end{bmatrix} = \begin{bmatrix} 2+0+0 & -2-1+0 & 2-2+0 \\ 3+0-8 & -3+1+12 & 3+2+4 \\ -2+0-6 & 2+5+9 & -2+10+3 \end{bmatrix} = \begin{bmatrix} 2 & -3 & 0 \\ -5 & 10 & 9 \\ -8 & 16 & 11 \end{bmatrix}$

3. Com relação às Matrizes do Exercício anterior. Calcule:
   a. Determinante da Matriz A (2,0)
   b. Matriz Inversa de B e C, caso existam. (2,0)
   c. Matriz transposta de A (2,0)

a) $A = \begin{bmatrix} 2 & -1 & 0 \\ 3 & 1 & 4 \\ -2 & 5 & 3 \end{bmatrix}$ det(A) = ?

$\begin{array}{ccc|cc} 2 & -1 & 0 & 2 & -1 \\ 3 & 1 & 4 & 3 & 1 \\ -2 & 5 & 3 & -2 & 5 \end{array}$

$2\times1\times3 = 6$
$-1\times4\times(-2) = 8$
$0\times3\times5 = 0$
$6+8 = 14$

$-1\times3\times3 = -9$
$2\times4\times5 = 2\times20 = 40$
$0\times1\times(-2) = 0$
$-9+40 = 31$

$\det(A) = 14-31 = -17$

b) $B = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ -2 & 3 & 1 \end{bmatrix}$ $B^{-1} = ?$

$\begin{bmatrix} 1 & -1 & 1 & 1 & -1 \\ 0 & 1 & 2 & 0 & 1 \\ -2 & 3 & 1 & -2 & 3 \end{bmatrix}$

$1\times1\times1 = 1$
$-1\times2\times(-2) = 4$
$1\times0\times3 = 0$
$1+4 = 5$

$-1\times0\times1 = 0$
$1\times2\times3 = 6$
$1\times1\times(-2) = -2$
$6-2 = 4$

$\det(B) = 5-4 = 1$

$1\times1-3\times2 = 1-6 = -5$
$3\times1-(-1)\times1 = 3+1 = 4$
$-1\times2-1\times1 = -2-1 = -3$

$2\times(-2)-1\times0 = -4$
$1\times1-1\times(-2) = 1-(-2) = 3$
$1\times0-2\times1 = 0-2 = -2$

$0\times3-(-2)\times1 = 2$
$-2\times(-1)-1\times3 = 2-3 = -1$
$1\times1-0\times(-1) = 1$

$\begin{bmatrix} -5 & -4 & 2 \\ 4 & 3 & -1 \\ -3 & -2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -5 & 4 & -3 \\ -4 & 3 & -2 \\ 2 & -1 & 1 \end{bmatrix}$

$B^{-1} = \frac{1}{\det(B)} \times \begin{bmatrix} -5 & 4 & -3 \\ -4 & 3 & -2 \\ 2 & -1 & 1 \end{bmatrix}$

$\frac{1}{1} = 1$ $B^{-1} = \begin{bmatrix} -5 & 4 & -3 \\ -4 & 3 & -2 \\ 2 & -1 & 1 \end{bmatrix}$

$C = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ -1 & 0 & 1 \end{bmatrix}$ det(C) = ?

$\begin{array}{ccc|cc} 2 & 1 & 2 & 2 & 1 \\ 4 & 2 & 4 & 4 & 2 \\ -1 & 0 & 1 & -1 & 0 \end{array}$

$2\times2\times1 = 4$
$1\times4\times(-1) = -4$
$2\times4\times0 = 0$
$4-4 = 0$

$1\times4\times1 = 4$
$2\times2\times(-1) = -4$
$4-4 = 0$

$0-0 = 0$

C Não tem inversa

c) $A = \begin{bmatrix} 2 & -1 & 0 \\ 3 & 1 & 4 \\ -2 & 5 & 3 \end{bmatrix}$

$A^T = \begin{bmatrix} 2 & 3 & -2 \\ -1 & 1 & 5 \\ 0 & 4 & 3 \end{bmatrix}$

4. Utilize o Método de Eliminação de Gauss para resolver o seguinte sistema: (2,0)

$$\begin{cases} x + 2y + z = 8 \\ 2x + y - z = 1 \\ -3x + y + z = -1 \end{cases}$$

$\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 2 & 1 & -1 & 1 \\ -3 & 1 & 1 & -1 \end{array} \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -3 & -3 & -15 \\ -3 & 1 & 1 & -1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -3 & -3 & -15 \\ 0 & 7 & 4 & 23 \end{array}\right] \rightarrow \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -3 & -3 & -15 \\ 0 & 0 & -3 & -12 \end{array}\right] \end{array}$

$L_2 = L_2 - 2 \times L_1$
$2-2\times1 = 0$
$1-2\times2 = 1-4 = -3$
$-1-2\times1 = -1-2 = -3$
$1-2\times8 = 1-16 = -15$

$L_3 = L_3 + 3 \times L_1$
$-3+3\times1 = -3+3 = 0$
$1+3\times2 = 1+6 = 7$
$1+3\times1 = 1+3 = 4$
$-1+3\times8 = -1+24 = 23$

$L_3 = L_3 + \frac{7}{3} \times L_2$
$7+\frac{7}{3}\times(-3) = 7+\left(-\frac{21}{3}\right) = 7-\frac{21}{3} = 7-7 = 0$
$4+\frac{7}{3}\times(-3) = 4-7 = -3$
$23+\frac{7}{3}\times(-15) = 23-\frac{105}{3} = \frac{69}{3}-\frac{105}{3} = \frac{-36}{3} = -12$

$$\begin{cases} 1x + 2y + 1z = 8 \\ -3y - 3z = -15 \\ -3z = -12 \end{cases}$$

$\frac{\cancel{-3}z}{\cancel{-3}} = \frac{-12}{-3}$
$z = \frac{-12}{-3}$
$z = \frac{12}{3}$
$\boxed{z = 4}$

$-3y - 3\times4 = -15$
$-3y - \cancel{12} = -15$
$+\cancel{12} \quad +12$
$-3y = -15+12$
$\frac{-3y}{-3} = \frac{-3}{-3}$
$y = \frac{-3}{-3}$
$y = \frac{3}{3}$
$\boxed{y = 1}$

$1x + 2\times1 + 1\times4 = 8$
$1x + 2 + 4 = 8$
$1x + \cancel{6} = 8$
$-\cancel{6} \quad -6$
$1x = 8-6$
$\boxed{x = 2}$

5. Sabendo que $T: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ é uma transformação linear e que $T(1,-1) = (3{,}2)$ e $T(-1{,}2) = (1,-1)$. Determine $T(x,y)$ (2,0)

$a\begin{pmatrix} 1 \\ -1 \end{pmatrix} + b\begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\begin{cases} 1a + (-1)b = x \\ -1a + 2b = y \end{cases}$$

$1a - 1b = x$
$a - \cancel{b} = x$
$+\cancel{b} \quad +b$
$a = x + b$

$-1a + 2b = y$
$-(x+b) + 2b = y$
$-x - b + 2b = y$
$-\cancel{x} + b = y$
$+\cancel{x} \quad +y$

$b = y + x$
$b: x + y$

$T(1,-1) = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$
$T(-1,2) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$a = x + x + y$
$a = 2x + y$ $b = x + y$

$2x+y\begin{bmatrix} 3 \\ 2 \end{bmatrix} + x+y\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

1ª Coordenada
$(2x+y)\times3 + (x+y)\times1$
$6x+3y + 1x + 1y \iff$
$7x + 4y \iff$

2ª Coordenada
$(2x+y)\times2 + (x+y)\times(-1)$
$4x + 2y - 1x - 1y$
$3x + 1y \iff$

$T(1,-1) = (7\times1 + 4\times(-1), 3\times1 + 1\times(-1)) = (7-4, 3-1) = (3,2)$ ✓
$T(-1,2) = (7\times(-1) + 4\times2, 3\times(-1) + 1\times2) = (-7+8, -3+2) = (1,-1)$ ✓

$T(x,y) = (7x + 4y, 3x + 1y)$

6. Verifique se os vetores $v_1 = (1{,}1{,}1) v_2 = (1{,}2{,}1) v_3 = (3{,}2,-1)$ são Linearmente Independentes (LI) no $\mathbb{R}^3$: (2,0)

$\vec{v_1} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ $\vec{v_2} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ $\vec{v_3} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$

$\left[\begin{array}{ccc|cc} 1 & 1 & 3 & 1 & 1 \\ 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & -1 & 1 & 1 \end{array}\right]$

$1\times2\times(-1) = -2$
$1\times2\times1 = 2$
$3\times1\times1 = 3$

$1\times1\times(-1) = -1$
$1\times2\times1 = 2$
$3\times2\times1 = 6$

$-2+2+3 = 3$
$-1+2+6 = 7$

$3-7 = -4$

$\det \neq 0 = LI$